

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Engineered and deployed **model serving architectures** on AWS, optimizing for **high throughput** and **low latency** across diverse environments, including **resource-constrained devices**.
- Developed and monitored **controlled inference tests** in simulated and live production environments, tracking key performance indicators such as **response latency**, **throughput**, and **memory consumption**.
- Analyzed computational efficiency and diagnosed bottlenecks in the serving pipeline by monitoring processing and memory metrics, addressing issues like suboptimal **batch processing** and **network delays**.
- Integrated optimized serving and inference frameworks into production pipelines for **edge and on-device applications**, defining success metrics for **real-world performance** and **scalability**.
- Designed and implemented **Python scripts** for data pipelines and ETL processes, supporting the development of **analytical dashboards** and operational reporting.
- Collaborated with cross-functional teams to gather requirements, conduct UAT, and perform post-implementation triage for enterprise business systems, including **ERP integrations** and operational platforms.

SKILLS

AI/ML – Model Serving, Inference Optimization, Kernel Optimization, Compute Shaders, MSL, AI R&D;, Machine Learning, NLP, Data Analytics, Statistical Analysis

Programming Languages – Python, Java, SQL, R, Metal Shading Language (MSL)

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn, Flask, FastAPI

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, Kubernetes

Data & BI – Power BI, Tableau, DAX, Power Query, Data Modelling, Data Warehousing, ETL

Development Tools – Git, Jenkins, Jira, Confluence, Splunk, Dynatrace

PROJECTS

Optimized Inference Pipeline for Edge Devices

- Developed a custom **inference pipeline** in Python and MSL for deployment on **resource-constrained edge devices**, focusing on **low-level kernel optimizations**.
- Implemented and evaluated **compute shaders** to accelerate model execution, reducing average inference latency by **38%** compared to baseline implementations.
- Established performance metrics including **memory footprint** and **throughput** under simulated network conditions, documenting findings for iterative refinement.

Data Analytics Dashboard for Financial Operations

- Created interactive **Power BI dashboards** to visualize key operational metrics from financial transaction data, enabling stakeholders to monitor **payment flows** and identify anomalies.
- Engineered data models and wrote complex **DAX measures** and **Power Query** scripts for data transformation and aggregation, handling approximately **1.5 million** transaction records weekly.
- Collaborated with business analysts to refine reporting requirements and ensure the dashboards provided actionable insights into operational efficiency and potential bottlenecks.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)